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Improving the Design and Implementation of Local and State Policies that Ban the Sale of Flavored Tobacco Products

Sarah D. Mills, PhD, MPH,[1,2] Frantasia Hill, MPH,[1] Clayton R. Ulm, BS,[1] Maya Wilson, MPH,[3] Evan Liebgott, MPH,[1]
Anne Donnelly, MPH,[1] Shelley D. Golden, PhD, MPH,[1,2] Kurt M. Ribisl, PhD,[1,2] Kristen Hassmiller Lich, PhD[3]

[1]Department of Health Behavior, UNC Gillings School of Global Public Health, [2]UNC Lineberger Comprehensive Cancer Center, [3]Department of Health Policy and Management, UNC Gillings School of Global Public Health

BACKGROUND

- Policies that ban the sale of flavored tobacco products are being implemented in a growing number of US cities and states
- Research is emerging about the impacts of flavor ban policies on tobacco use behavior
- Contextual factors may play an important role in the effectiveness of flavor bans

OBJECTIVES

1. Conduct a systematic review and meta-analysis of studies examining the impacts of flavored cigar bans
2. Develop a causal loop diagram to visualize contextual factors that may impact ban effectiveness

METHODS

Systematic Review & Meta Analysis

- Searched PubMed, PsycINFO, CINAHL, and ASSIA
- Included studies on impacts of real-world or hypothesized flavored cigar bans on tobacco use
  - Secondary outcomes: retailer compliance, sales, tobacco industry response
- Two reviewers independently screened articles, extracted data, and assessed study quality
- Conducted a meta-analysis to pool estimates on tobacco use behavior

Causal Loop Diagram

- Conducted the review above and updated a systematic review of studies on menthol bans
- Two researchers coded results sections of studies, extracting cause-and-effect relationships among variables related to flavor ban effectiveness
- Cause-and-effect relationships were synthesized in a causal loop diagram (CLD) using Kumu software

RESULTS

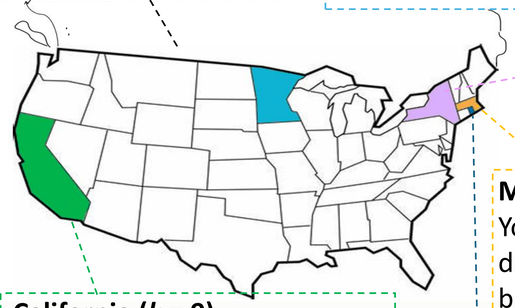
Systematic Review & Meta-Analysis

Real-world Behavior

When sales of flavored cigars were banned, what did people do?

Canada (*k* = 1)
Cigarillo use and all cigar use decreased among 15-24-year-olds

Minneapolis and Saint Paul (*k* = 2)
Youth cigar use did not decline until menthol was added to the ban



New York City (*k* = 1)
Youth cigarillo use did not change

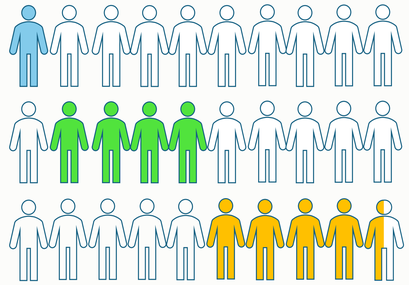
Massachusetts (*k* = 5)
Youth flavored cigar use decreased in one city with a ban, but not in others

California (*k* = 9)
Flavored and any cigar use declined among 18-24- year-olds, but no change among 25-35-year-olds

Providence (*k* = 1)
Cigar use among high schoolers decreased

Intended Behavior (*k* = 11)

If sales of flavored cigars are banned, what would you do?



8% would quit all tobacco products

41% would switch to other tobacco products

45% would switch to non-flavored cigars or cigarettes

Secondary Outcomes

- *Retailer compliance* ranged from 29% - 100%
- *Sales* of flavored cigars decreased, but sales of non-flavored tobacco products increased in some places
- The *tobacco industry* increased the weight of cigars and introduced new concept-flavored cigarillos

Causal Loop Diagram

The complete CLD contains 40 constructs, 92 connections (i.e., arrows) linking constructs together, and 23 feedback loops. We identified four key narratives: 1) motivations for a menthol ban, 2) tobacco industry strategies to circumvent or delay bans, 3) factors affecting retailer compliance; and 4) the effects of a menthol ban over time. Below we focus on two narratives.

Figure 1. Tobacco Industry Tactics

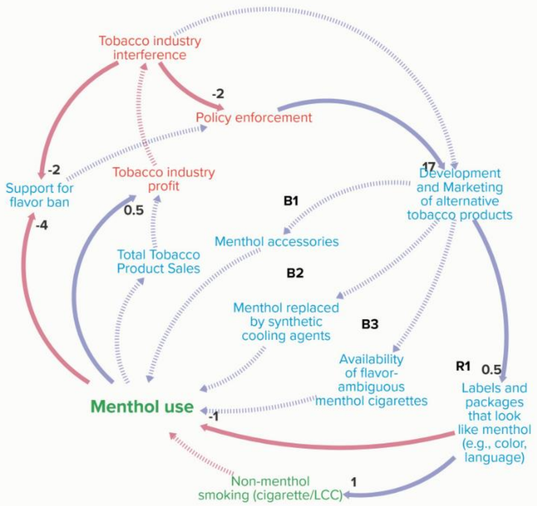
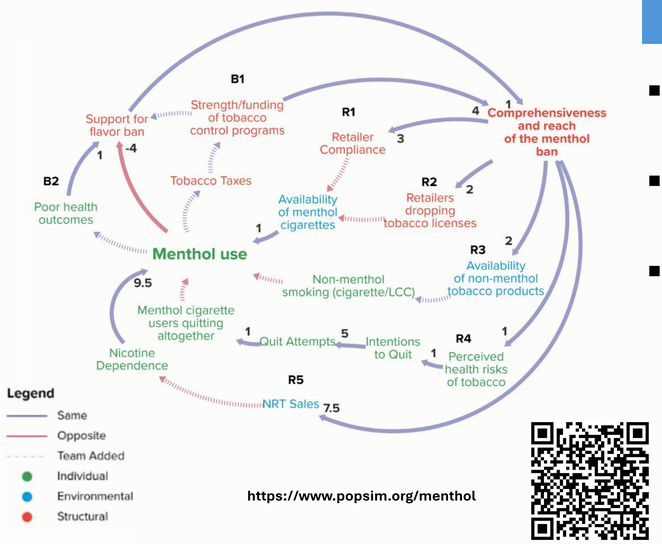


Figure 2. Impacts of a Menthol Ban



CONCLUSIONS

- Most studies find a decrease in cigar use, but others do not
- People intend to switch tobacco products instead of quitting
- CLD suggests: 1) industry tactics to circumvent the ban must be mitigated, 2) comprehensive bans are more effective, and 3) cessation support should be expanded